Ubuntu Web Stack Setup & Security Notes

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1. Initial System Setup

```
sudo apt update && sudo apt upgrade -y

sudo apt install curl wget unzip git ufw net-tools htop vim -y
```

2. Apache Installation & Configuration

```
sudo apt install apache2 -y

sudo systemctl enable apache2
sudo systemctl start apache2

systemctl status apache2
```

Default web root:
/var/www/html

Apache configuration directories:
/etc/apache2/
/etc/apache2/sites-available/
/etc/apache2/sites-enabled/

3. Virtual Hosts (Virtual Mapping)

```
example.local → /var/www/example
api.local     → reverse proxy to Python app
```

```
sudo mkdir -p /var/www/example
sudo mkdir -p /var/www/api

sudo chown -R $USER:$USER /var/www/example
sudo chown -R $USER:$USER /var/www/api

echo "<h1>Example Site</h1>" > /var/www/example/index.html
echo "<h1>API Site</h1>" > /var/www/api/index.html
```

4. Apache Virtual Host Configs

example.conf:
/etc/apache2/sites-available/example.conf

```
<VirtualHost *:80>
    ServerName example.local
    DocumentRoot /var/www/example

    <Directory /var/www/example>
        AllowOverride All
        Require all granted
        Options -Indexes
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/example_error.log
    CustomLog ${APACHE_LOG_DIR}/example_access.log combined
</VirtualHost>
```

```
sudo a2ensite example.conf
sudo systemctl reload apache2
```

api.conf:
/etc/apache2/sites-available/api.conf

```
<VirtualHost *:80>
    ServerName api.local

    ProxyPass / http://127.0.0.1:8000/
    ProxyPassReverse / http://127.0.0.1:8000/

    ErrorLog ${APACHE_LOG_DIR}/api_error.log
    CustomLog ${APACHE_LOG_DIR}/api_access.log combined
</VirtualHost>
```

```
sudo a2enmod proxy proxy_http
```

```
sudo a2ensite api.conf
sudo systemctl reload apache2
```

5. Local Domain Mapping

```
sudo nano /etc/hosts
```

```
127.0.0.1 example.local
127.0.0.1 api.local
```

6. PHP Installation

```
sudo apt install php libapache2-mod-php php-mysql php-redis php-cli php-curl php-xml php-mbstring -y
```

```
echo "<?php phpinfo(); ?>" | sudo tee /var/www/html/info.php
```

7. MySQL Installation & Security

```
sudo apt install mysql-server -y
```

```
sudo mysql_secure_installation
```

```
/etc/mysql/mysql.conf.d/mysqld.cnf
```

```
bind-address = 127.0.0.1
```

```
sudo mysql
```

```
CREATE DATABASE appdb;
```

```
CREATE USER 'appuser'@'localhost' IDENTIFIED BY 'StrongPassword';
```

```
GRANT ALL PRIVILEGES ON appdb.* TO 'appuser'@'localhost';
FLUSH PRIVILEGES;
```

8. Redis Installation & Hardening

```
sudo apt install redis-server -y
```

```
sudo systemctl enable redis-server
sudo systemctl start redis-server
```

```
redis-cli ping
```

Expected:
PONG

```
/etc/redis/redis.conf
```

```
bind 127.0.0.1
protected-mode yes
requirepass StrongRedisPassword
```

```
sudo systemctl restart redis-server
```

```
redis-cli
AUTH StrongRedisPassword
```

9. Python Setup

```
sudo apt install python3 python3-pip python3-venv -y
```

```
sudo apt install python-is-python3
```

```
python3 -m venv venv
source venv/bin/activate
```

```
pip install flask
```

10. Apache Reverse Proxy for Python

```
Apache :80
    ↓
Gunicorn/Flask :8000
```

```
sudo a2enmod proxy proxy_http
sudo systemctl restart apache2
```

11. Firewall (UFW)

```
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp

sudo ufw enable

sudo ufw status
```

12. Apache Security Hardening

```
sudo nano /etc/apache2/conf-enabled/security.conf

ServerTokens Prod
ServerSignature Off
TraceEnable Off
```

13. Disable Directory Listing

```
Options -Indexes
```

14. Useful Apache Modules

```
sudo a2enmod rewrite headers ssl proxy proxy_http
```

15. SSH Hardening

```
sudo nano /etc/ssh/sshd_config

PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes

sudo systemctl restart ssh

groups
```

Look for:
sudo

16. Disk Encryption Decision

Skipped full disk encryption for:
- Learning labs
- Local VMs
- Lower CPU usage

Tradeoff:
- Better performance
- No at-rest encryption

17. Service Management

```
systemctl status apache2
systemctl status mysql
systemctl status redis-server

sudo systemctl restart apache2

sudo systemctl enable apache2
```

18. Checking Open Ports

```
sudo ss -tulpn
```

19. Logs

Apache:
/var/log/apache2/access.log
/var/log/apache2/error.log

```
sudo tail -f /var/log/apache2/error.log
```

MySQL:
sudo journalctl -u mysql

Redis:
sudo journalctl -u redis-server

20. Certbot / Let's Encrypt Note

```
sudo apt install certbot python3-certbot-apache -y
```

Let's Encrypt does NOT issue certificates for local/fake domains:
- example.local
- api.local

Use HTTP locally or self-signed certificates for testing.